UPiS - UPS for Raspberry Pi 5

PCB Project:	UPiS Motherboard
Version:	VER2
Revision:	REV1
Project State:	Released (2026-02-06)
Variant:	[No Variations]
Print date:	3/9/2026

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02	PD Source
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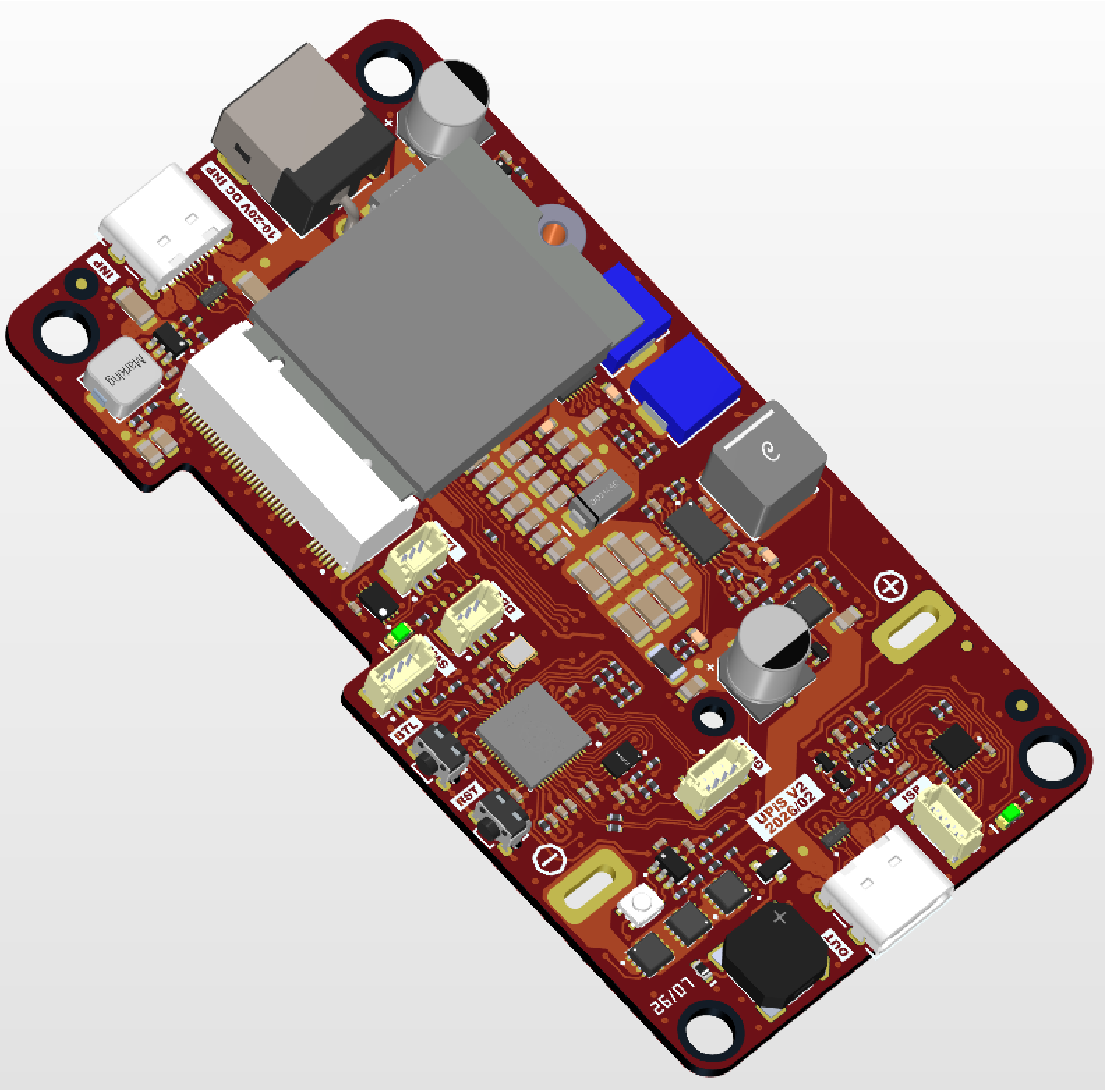
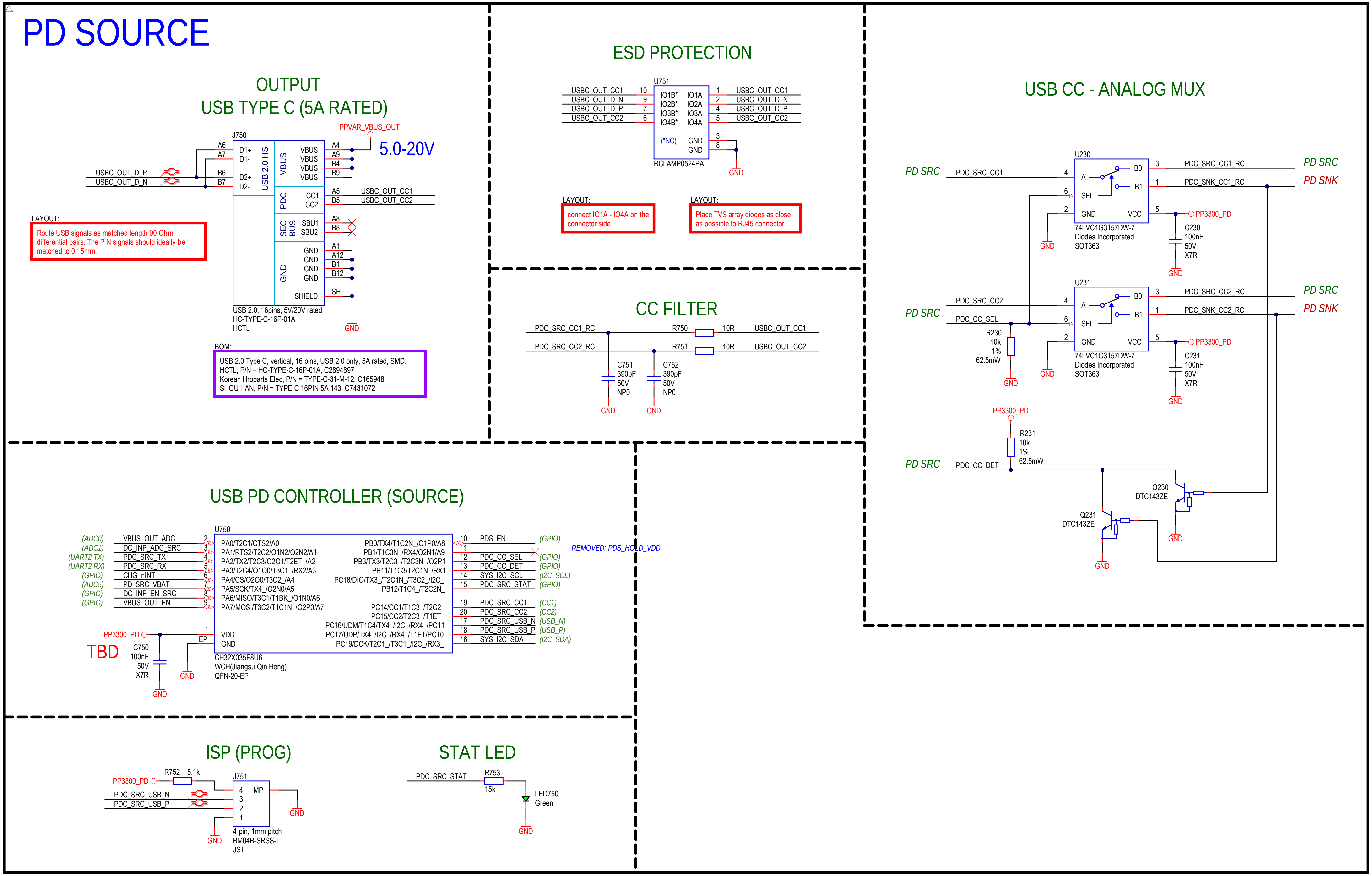
05	Charger
06	PD Buck Boost Regulator (OUT)
07	SYS I2C & UART
08	RP2040
09	UI
10	ULP (Ultra Low Power management)
11	PCB Marking
12	NGFF

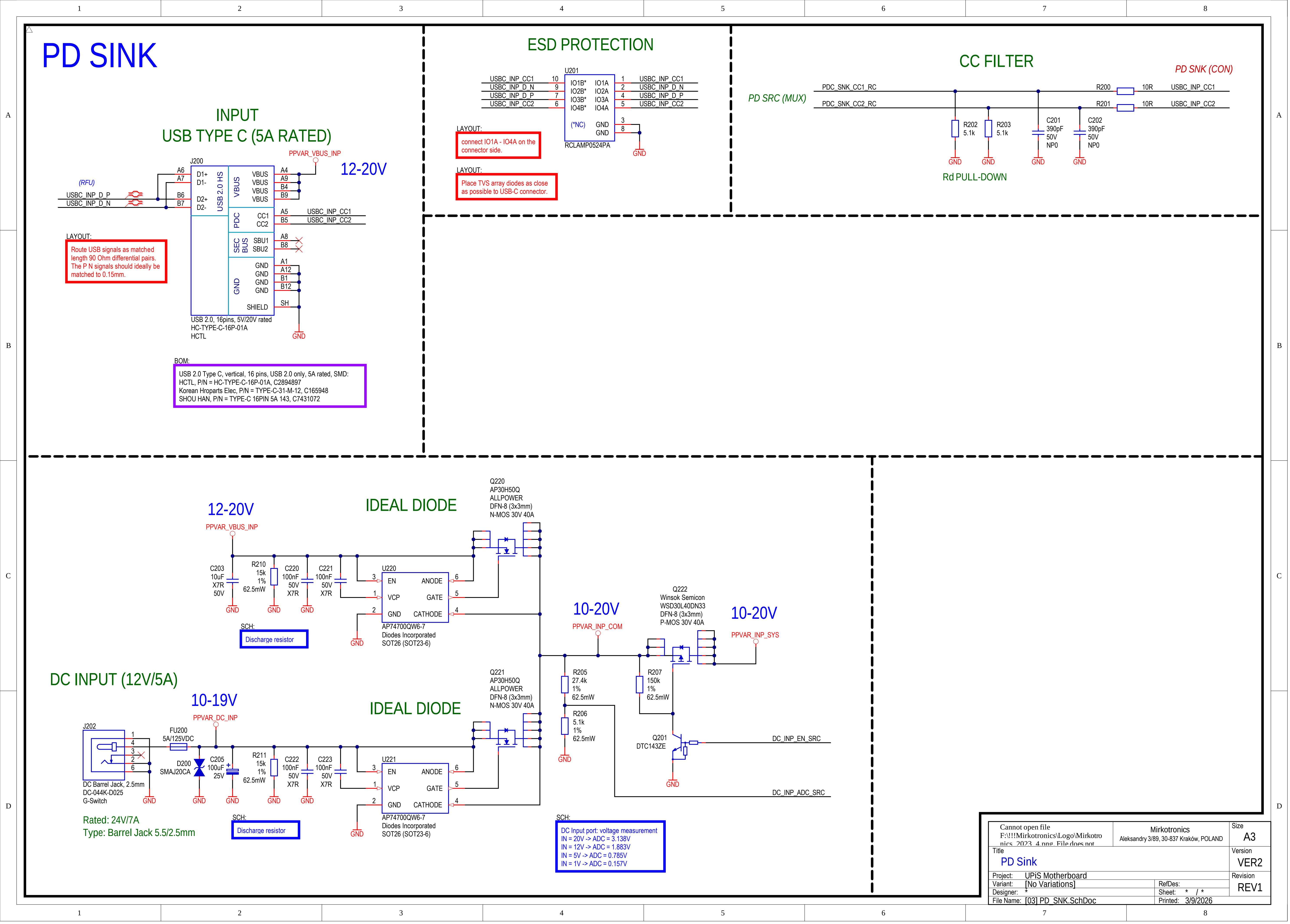
PCB VIEW



[02] PD_SRC.SchDoc
[03] PD_SNK.SchDoc
[04] PCM.SchDoc
[05] Charger.SchDoc
[06] PD_BB_Out.SchDoc
[07] SYS_I2C_UART.SchDoc
[08] RP2040.SchDoc
[09] UI.SchDoc
[10] ULP.SchDoc
[11] PCB_Marking.SchDoc
[12] NGFF.SchDoc

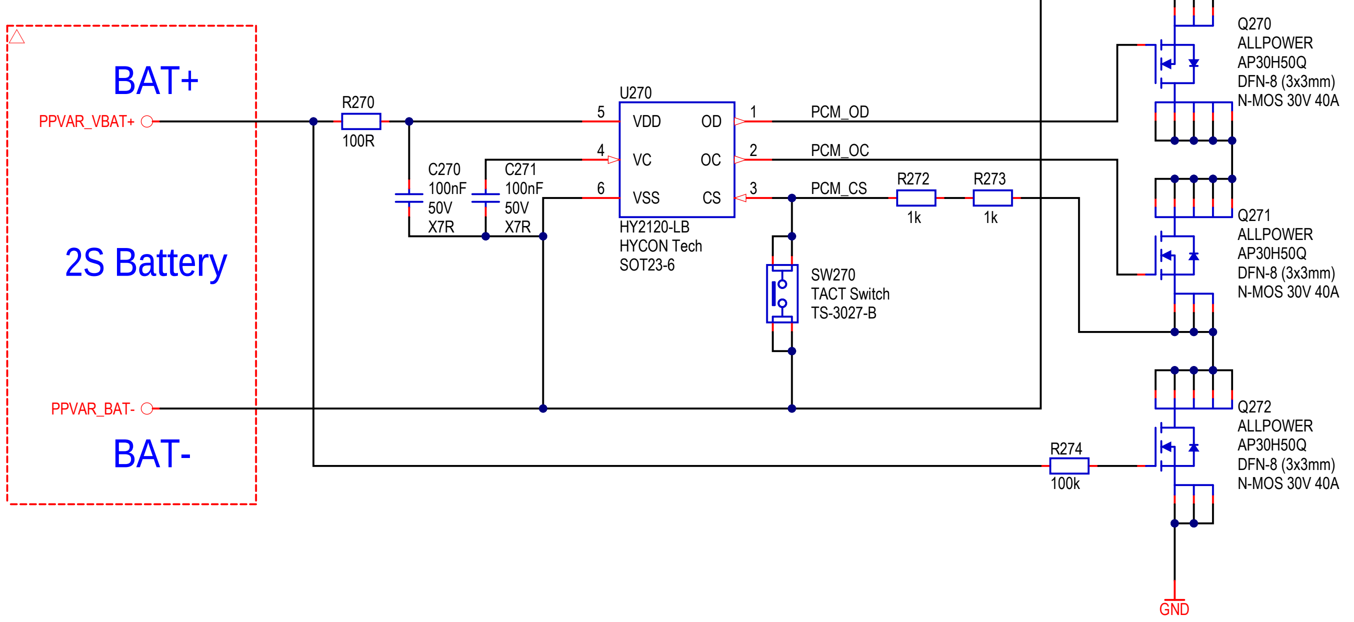


Cannot open file F:\!!!Mirkotronics\Logo\Mirkotronics_2023_4.png. File does not	Mirkotronics Aleksandry 3/89, 30-837 Kraków, POLAND		Size A3
Title PD Source			Version VER2
Project: UPiS Motherboard			Revision REV1
Variant: [No Variations]		RefDes:	
Designer: *		Sheet: * / *	
File Name: [02] PD_SRC.SchDoc		Printed: 3/9/2026	


PD SINK
INPUT
USB TYPE C (5A RATED)
J200
PPVAR_VBUS_INP
12-20V
(RFU)
USBC_INP_D_P
USBC_INP_D_N
D1+
D1-
D2+
D2-
USB 2.0 HS
VBUS
PDC
SEC BUS
GND
CC1
CC2
SBU1
SBU2
SHIELD
USBC_INP_CC1
USBC_INP_CC2
USB 2.0, 16pins, 5V/20V rated
HC-TYPE-C-16P-01A
HCTL
LAYOUT:
Route USB signals as matched length 90 Ohm differential pairs. The P N signals should ideally be matched to 0.15mm.
BOM:
USB 2.0 Type C, vertical, 16 pins, USB 2.0 only, 5A rated, SMD:
HCTL, P/N = HC-TYPE-C-16P-01A, C2894897
Korean Hroparts Elec, P/N = TYPE-C-31-M-12, C165948
SHOU HAN, P/N = TYPE-C 16PIN 5A 143, C7431072
ESD PROTECTION
U201
RCLAMP0524PA
(*NC)
LAYOUT:
connect IO1A - IO4A on the connector side.
LAYOUT:
Place TVS array diodes as close as possible to USB-C connector.
CC FILTER
PD SNK (CON)
PD SRC (MUX)
PDC_SNK_CC1_RC
PDC_SNK_CC2_RC
R200 10R
R201 10R
R202 5.1k
R203 5.1k
C201 390pF 50V NP0
C202 390pF 50V NP0
Rd PULL-DOWN
12-20V
PPVAR_VBUS_INP
IDEAL DIODE
Q220
AP30H50Q
ALLPOWER
DFN-8 (3x3mm)
N-MOS 30V 40A
C203 10uF X7R 50V
R210 15k 1% 62.5mW
C220 100nF 50V X7R
C221 100nF 50V X7R
U220
EN
VCP
GND
ANODE
GATE
CATHODE
AP74700QW6-7
Diodes Incorporated
SOT26 (SOT23-6)
SCH:
Discharge resistor
10-20V
PPVAR_INP_COM
Q222
Winsok Semicon
WSD30L40DN33
DFN-8 (3x3mm)
P-MOS 30V 40A
10-20V
PPVAR_INP_SYS
R205 27.4k 1% 62.5mW
R206 5.1k 1% 62.5mW
R207 150k 1% 62.5mW
Q201
DTC143ZE
DC_INP_EN_SRC
DC_INP_ADC_SRC
DC INPUT (12V/5A)
10-19V
PPVAR_DC_INP
J202
FU200
5A/125VDC
D200
SMAJ20CA
C205 100uF 25V
R211 15k 1% 62.5mW
C222 100nF 50V X7R
C223 100nF 50V X7R
DC Barrel Jack, 2.5mm
DC-044K-D025
G-Switch
Rated: 24V/7A
Type: Barrel Jack 5.5/2.5mm
IDEAL DIODE
Q221
AP30H50Q
ALLPOWER
DFN-8 (3x3mm)
N-MOS 30V 40A
U221
AP74700QW6-7
Diodes Incorporated
SOT26 (SOT23-6)
SCH:
Discharge resistor
SCH:
DC Input port: voltage measurement
IN = 20V -> ADC = 3.138V
IN = 12V -> ADC = 1.883V
IN = 5V -> ADC = 0.785V
IN = 1V -> ADC = 0.157V
Mirkotronics
Aleksandry 3/89, 30-837 Kraków, POLAND
Title
PD Sink
Size A3
Version VER2
Revision REV1
Project: UPiS Motherboard
Variant: [No Variations]
Designer: *
RefDes:
Sheet: * / *
File Name: [03] PD_SNK.SchDoc
Printed: 3/9/2026

PCM

BATTERY PROTECTION (PCM)



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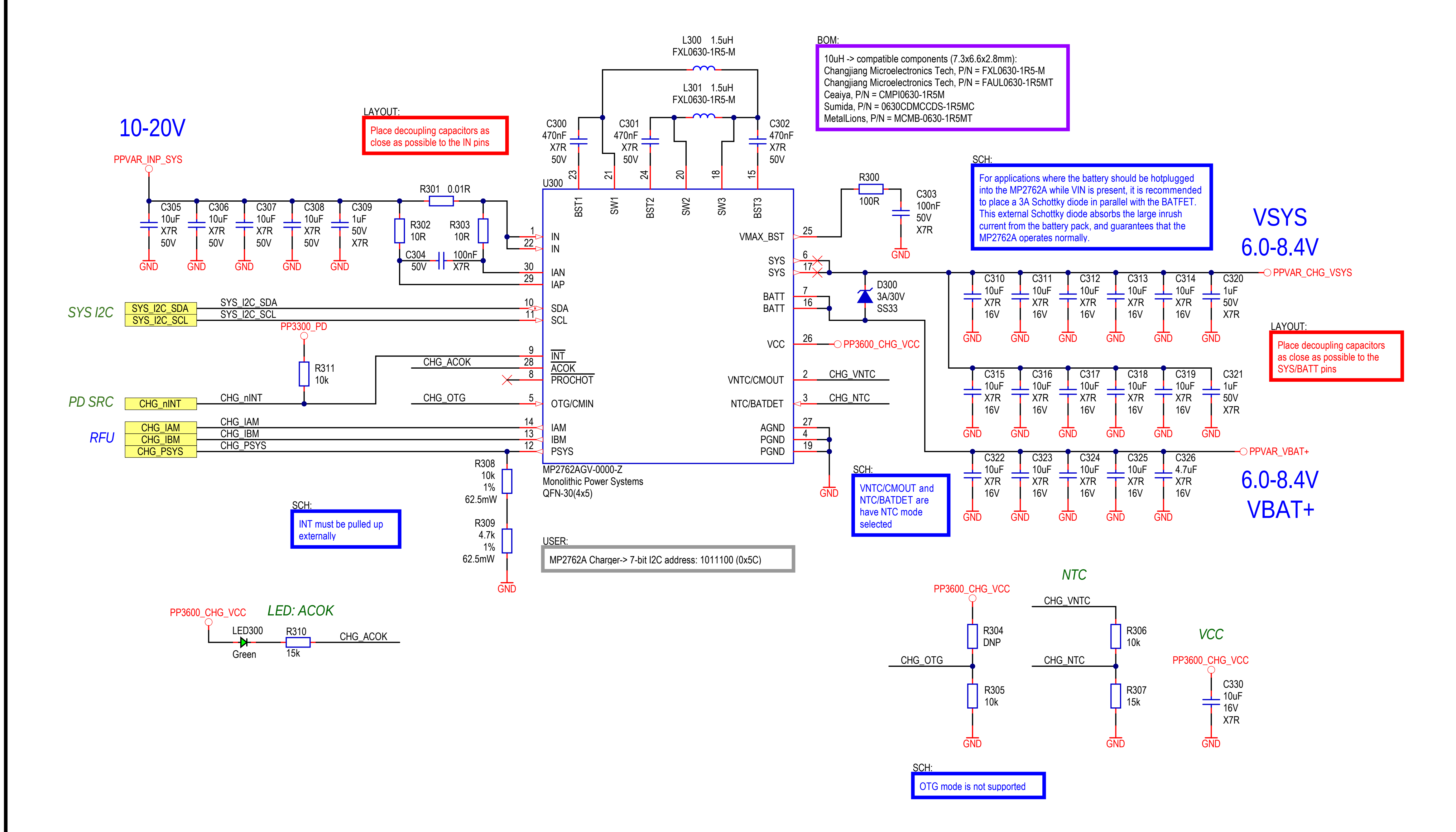
Title PCM		Version VER2
Project: UPiS Motherboard Variant: [No Variations] Designer: * File Name: [04] PCM.SchDoc	RefDes: Sheet: * / * Printed: 3/9/2026	Revision REV1

CHARGER

6A/20Vin 2S Li-Ion / Li-Pol Charger



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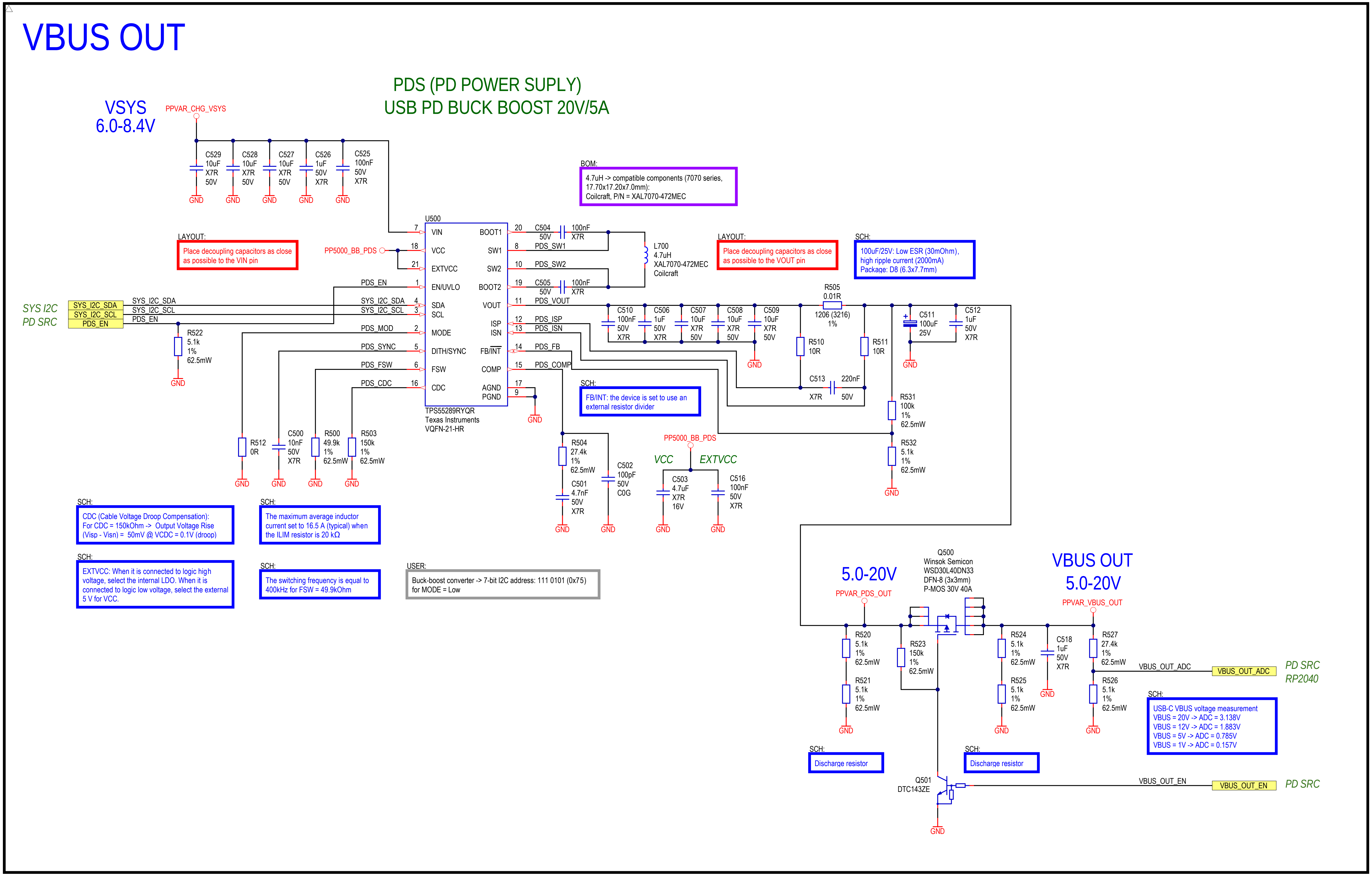
Title: 6A, 2S Li-Ion/Li-Pol Buck-boost Charger			Version VER2
Project: UPiS Motherboard			Revision REV1
Variant: [No Variations]		RefDes:	
Designer: *		Sheet: * / *	
File Name: [05] Charger.SchDoc		Printed: 3/9/2026	



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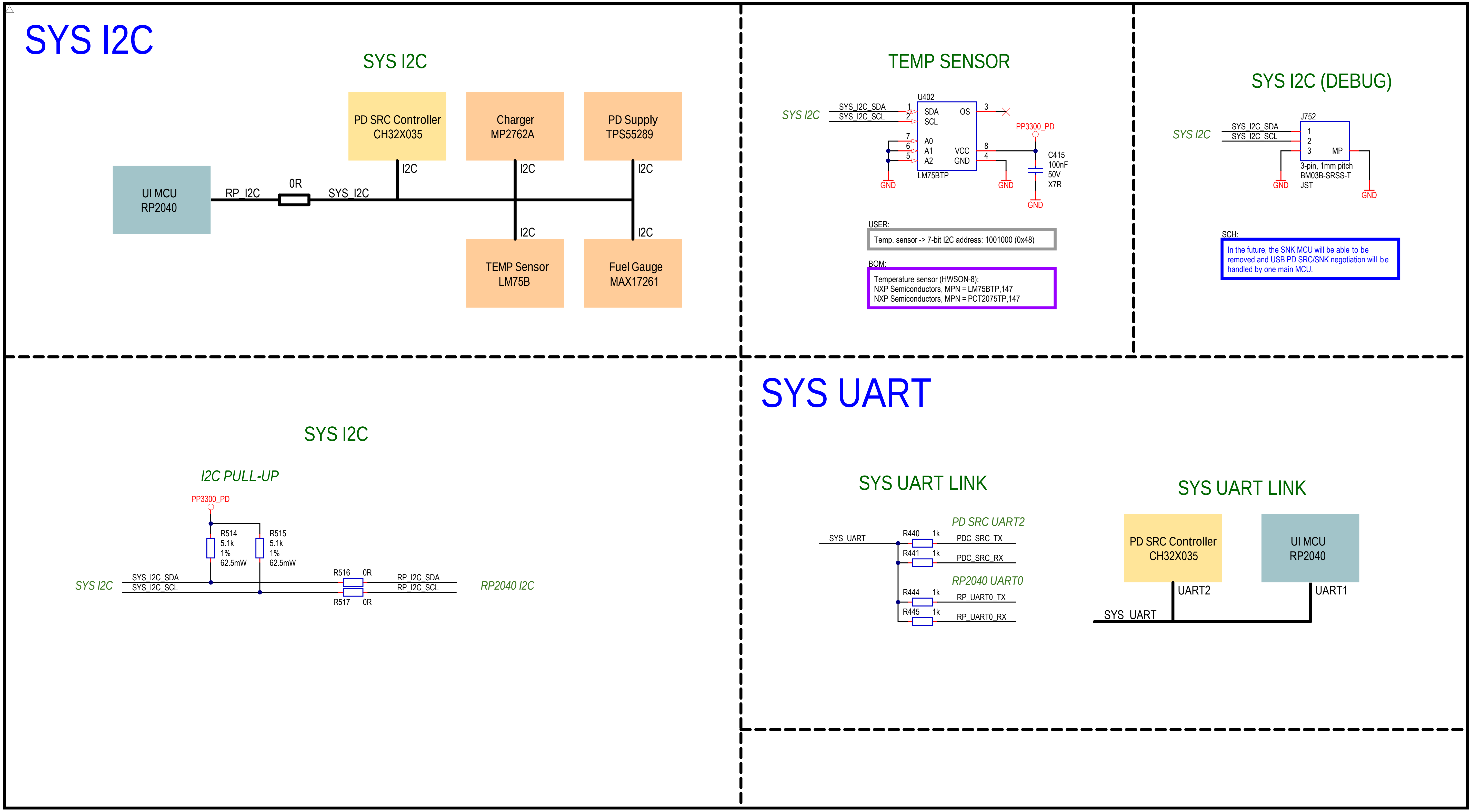
Size	A3
Title	VBUS Output - Buck-boost regulator
Version	VER2
Project:	UPiS Motherboard
Variant:	[No Variations]
RefDes:	
Revision	REV1
Designer:	*
Sheet:	* / *
File Name:	[06] PD_BB_Out.SchDoc
Printed:	3/9/2026



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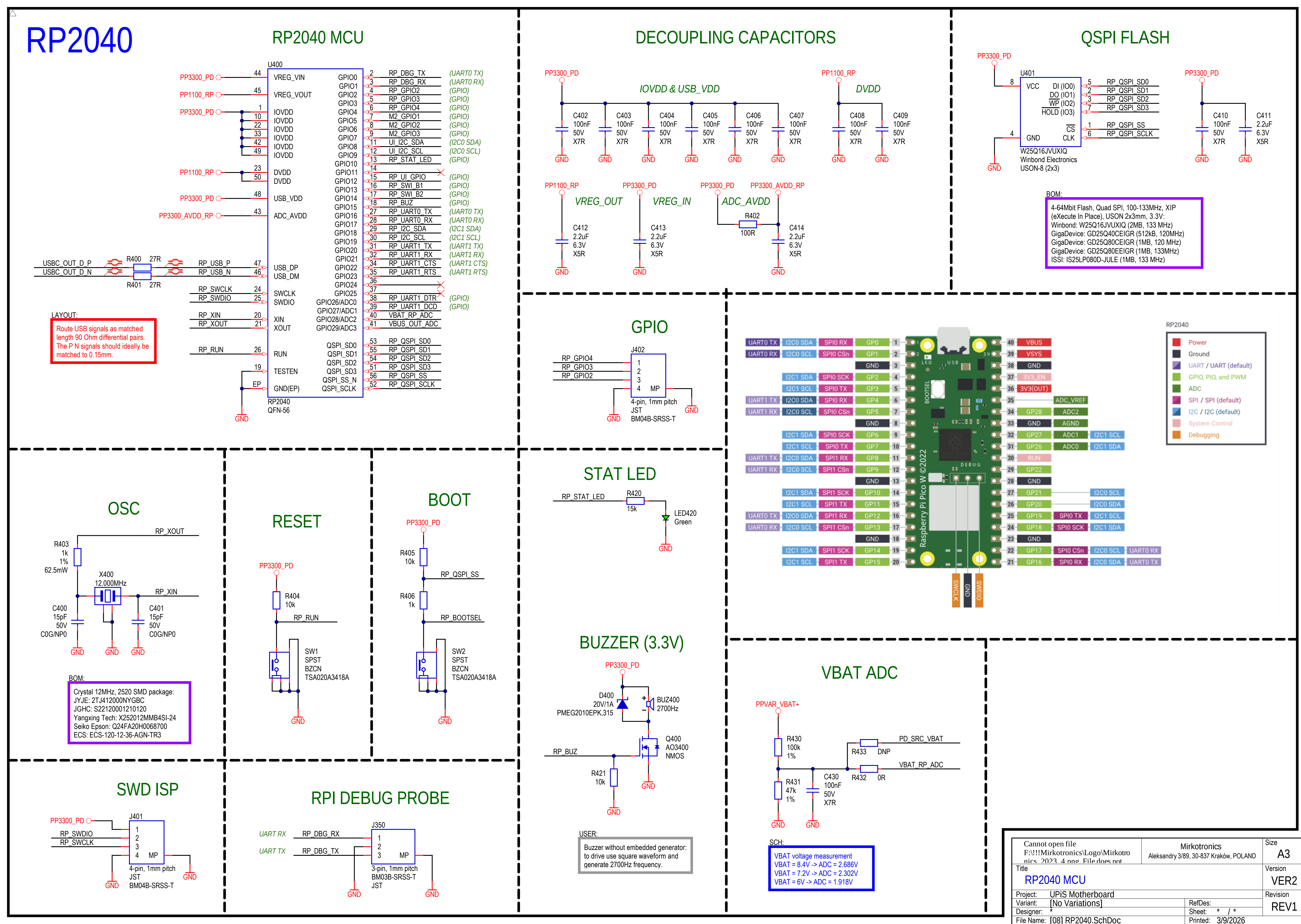
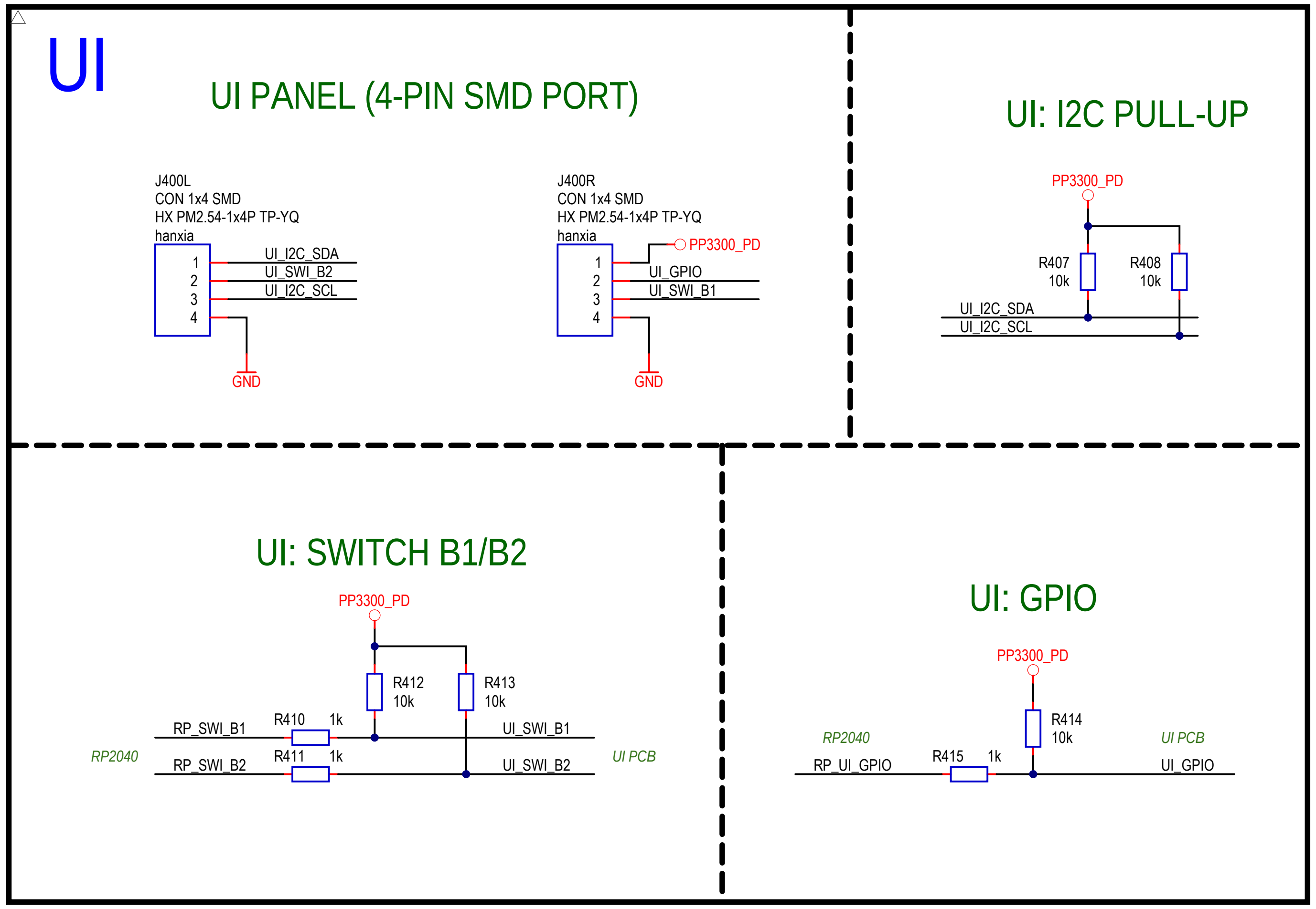
Title SYS I2C & UART		Version VER2
Project: UPiS Motherboard Variant: [No Variations] Designer: * File Name: [07] SYS_I2C_UART.SchDoc	RefDes: Sheet: * / * Printed: 3/9/2026	Revision REV1


RP2040
RP2040 MCU
U400
VREG_VIN
VREG_VOUT
IOVDD
DVDD
USB_VDD
ADC_AVDD
USB_DP
USB_DM
SWCLK
SWDIO
XIN
XOUT
RUN
TESTEN
GND(EP)
RP2040
QFN-56
PP3300_PD
PP1100_RP
PP3300_AVDD_RP
USBC_OUT_D_P
USBC_OUT_D_N
R400 27R
R401 27R
RP_USB_P
RP_USB_N
RP_SWCLK
RP_SWDIO
RP_XIN
RP_XOUT
RP_RUN
RP_DBG_TX (UART0 TX)
RP_DBG_RX (UART0 RX)
RP_GPIO2 (GPIO)
RP_GPIO3 (GPIO)
RP_GPIO4 (GPIO)
M2_GPIO1 (GPIO)
M2_GPIO2 (GPIO)
M2_GPIO3 (GPIO)
UI_I2C_SDA (I2C0 SDA)
UI_I2C_SCL (I2C0 SCL)
RP_STAT_LED (GPIO)
RP_UI_GPIO (GPIO)
RP_SWI_B1 (GPIO)
RP_SWI_B2 (GPIO)
RP_BUZ (GPIO)
RP_UART0_TX (UART0 TX)
RP_UART0_RX (UART0 RX)
RP_I2C_SDA (I2C1 SDA)
RP_I2C_SCL (I2C1 SCL)
RP_UART1_TX (UART1 TX)
RP_UART1_RX (UART1 RX)
RP_UART1_CTS (UART1 CTS)
RP_UART1_RTS (UART1 RTS)
RP_UART1_DTR (GPIO)
RP_UART1_DCD (GPIO)
VBAT_RP_ADC
VBUS_OUT_ADC
RP_QSPI_SD0
RP_QSPI_SD1
RP_QSPI_SD2
RP_QSPI_SD3
RP_QSPI_SS
RP_QSPI_SCLK
LAYOUT:
Route USB signals as matched length 90 Ohm differential pairs. The P N signals should ideally be matched to 0.15mm.
DECOUPLING CAPACITORS
IOVDD & USB_VDD
DVDD
C402 C403 C404 C405 C406 C407 100nF 50V X7R
C408 C409 100nF 50V X7R
VREG_OUT
VREG_IN
ADC_AVDD
C412 2.2uF 6.3V X5R
C413 2.2uF 6.3V X5R
R402 100R
C414 2.2uF 6.3V X5R
QSPI FLASH
U401
W25Q16JVUXIQ
Winbond Electronics
USON-8 (2x3)
C410 100nF 50V X7R
C411 2.2uF 6.3V X5R
BOM:
4-64Mbit Flash, Quad SPI, 100-133MHz, XIP (eXecute In Place), USON 2x3mm, 3.3V:
Winbond: W25Q16JVUXIQ (2MB, 133 MHz)
GigaDevice: GD25Q40CEIGR (512kB, 120MHz)
GigaDevice: GD25Q80CEIGR (1MB, 120 MHz)
GigaDevice: GD25Q80EEIGR (1MB, 133MHz)
ISSI: IS25LP080D-JULE (1MB, 133 MHz)
GPIO
J402
4-pin, 1mm pitch
JST
BM04B-SRSS-T
RP2040
Power
Ground
UART / UART (default)
GPIO, PIO, and PWM
ADC
SPI / SPI (default)
I2C / I2C (default)
System Control
Debugging
Raspberry Pi Pico W ©2022
STAT LED
R420 15k
LED420 Green
OSC
R403 1k 1% 62.5mW
X400 12.000MHz
C400 15pF 50V C0G/NP0
C401 15pF 50V C0G/NP0
BOM:
Crystal 12MHz, 2520 SMD package:
JYJE: 2TJ412000NYGBC
JGHC: S22120001210120
Yangxing Tech: X252012MMB4SI-24
Seiko Epson: Q24FA20H0068700
ECS: ECS-120-12-36-AGN-TR3
RESET
R404 10k
SW1 SPST BZCN TSA020A3418A
BOOT
R405 10k
R406 1k
RP_BOOTSEL
SW2 SPST BZCN TSA020A3418A
BUZZER (3.3V)
D400 20V/1A PMEG2010EPK,315
BUZ400 2700Hz
Q400 AO3400 NMOS
R421 10k
USER:
Buzzer without embedded generator: to drive use square waveform and generate 2700Hz frequency.
VBAT ADC
PPVAR_VBAT+
R430 100k 1%
R431 47k 1%
C430 100nF 50V X7R
R433 DNP
R432 0R
PD_SRC_VBAT
VBAT_RP_ADC
SCH:
VBAT voltage measurement
VBAT = 8.4V -> ADC = 2.686V
VBAT = 7.2V -> ADC = 2.302V
VBAT = 6V -> ADC = 1.918V
SWD ISP
J401
4-pin, 1mm pitch
JST
BM04B-SRSS-T
RPI DEBUG PROBE
UART RX
UART TX
J350
3-pin, 1mm pitch
BM03B-SRSS-T
JST
Mirkotronics
Aleksandry 3/89, 30-837 Kraków, POLAND
Size A3
Title RP2040 MCU
Version VER2
Project: UPiS Motherboard
Variant: [No Variations]
Designer: *
File Name: [08] RP2040.SchDoc
RefDes:
Sheet: * / *
Printed: 3/9/2026
Revision REV1



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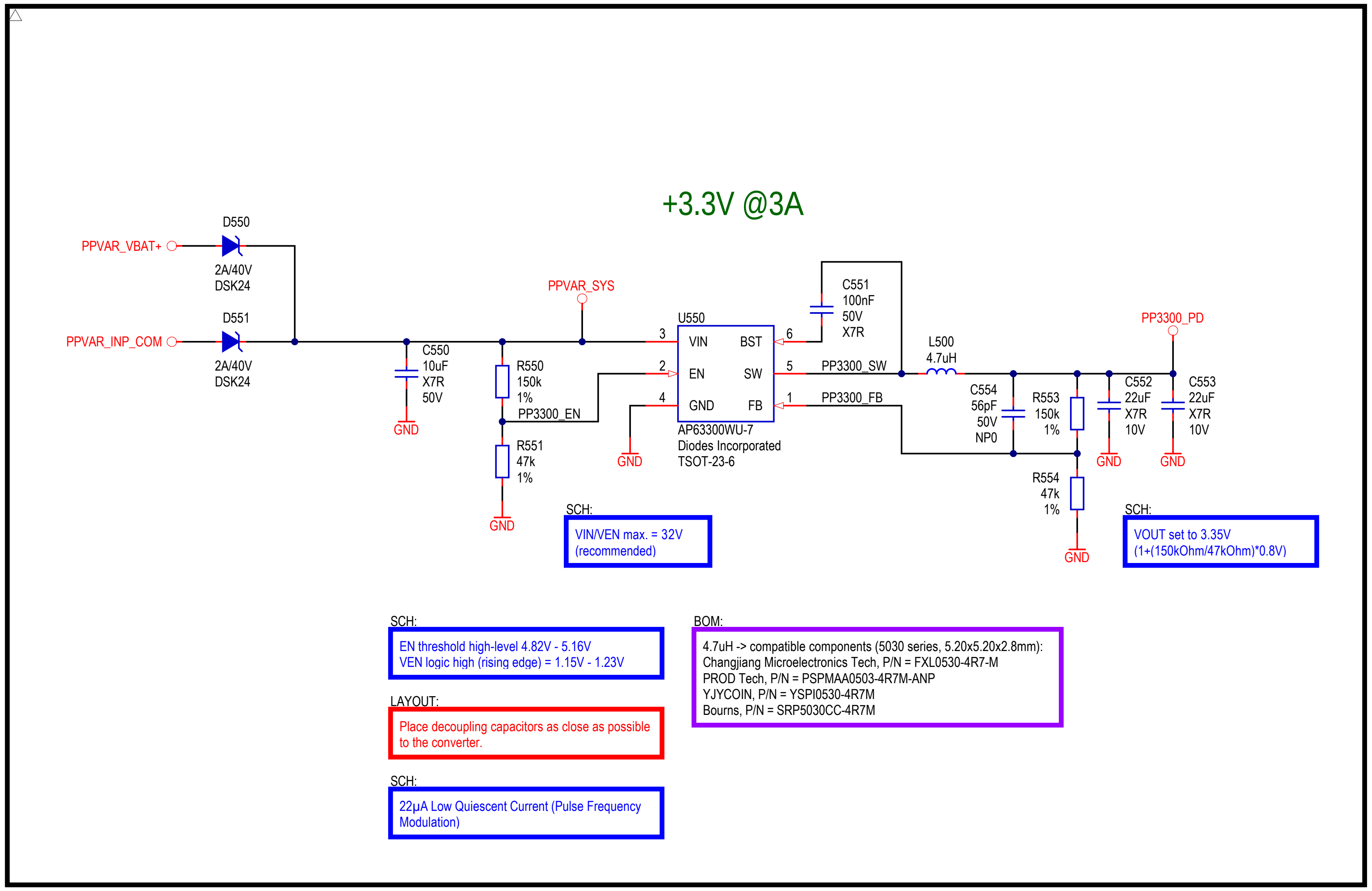
Mirkotronics
Aleksandry 3/89, 30-837 Kraków, POLAND

Size	A3
Title	UI - Front Panel
Version	VER2
Project:	UPiS Motherboard
Variant:	[No Variations]
Designer:	*
File Name:	[09] UI.SchDoc
RefDes:	
Sheet:	* / *
Printed:	3/9/2026
Revision	REV1

ULP



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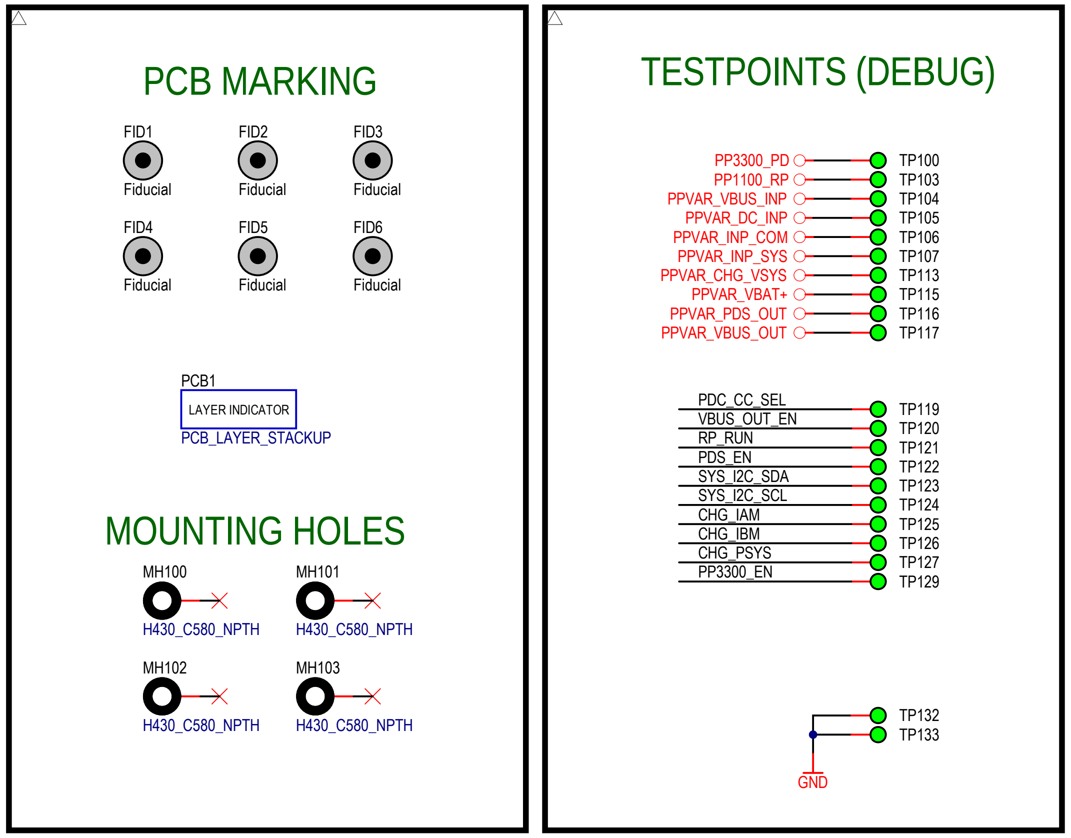
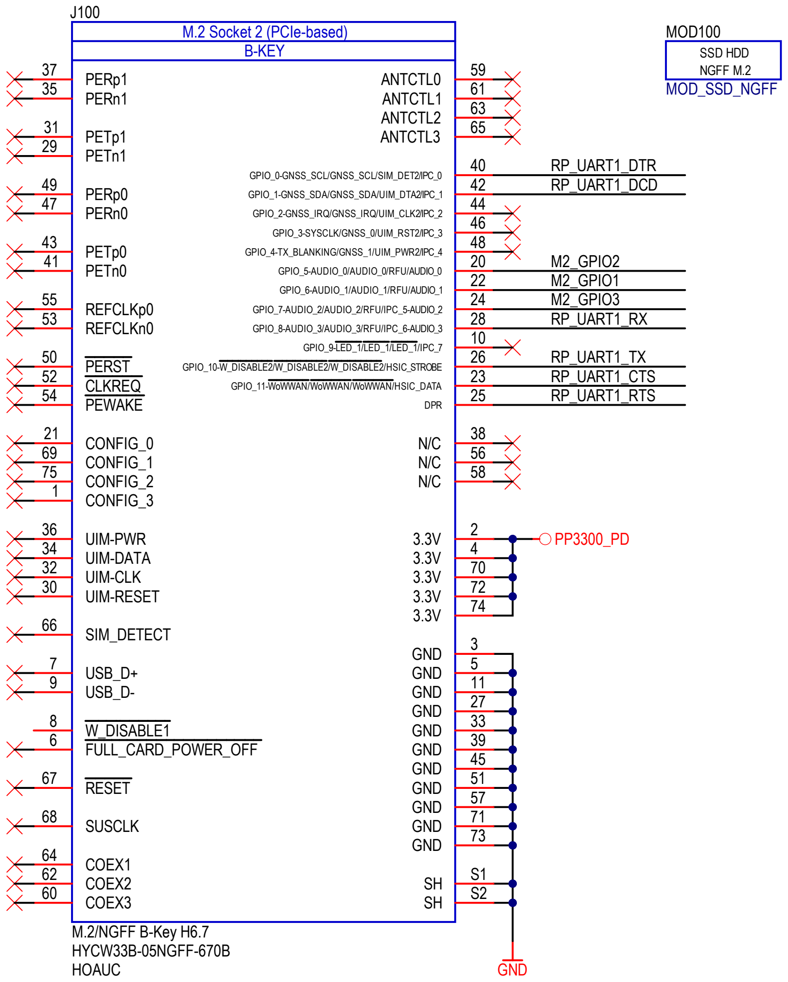
Mirkotronics
Aleksandry 3/89, 30-837 Kraków, POLAND

Size	A3
Title	ULP (Ultra Low Power management)
Version	VER2
Project:	UPiS Motherboard
Variant:	[No Variations]
Designer:	*
File Name:	[10] ULP.SchDoc
RefDes:	
Sheet:	* / *
Printed:	3/9/2026
Revision	REV1



Cannot open file F:\!!!Mirkotronics\Logo\Mirkotronics_2023_4.png. File does not	Mirkotronics Aleksandry 3/89, 30-837 Kraków, POLAND	Size A3
Title: PCB Marking & Testpoints		Version VER2
Project: UPiS Motherboard		Revision REV1
Variant: [No Variations]	RefDes:	
Designer: *	Sheet: * / *	
File Name: [11] PCB_Marking.SchDoc	Printed: 3/9/2026	

WWAN Module - M.2 Socket 2 (B-Key) PCIe-Based



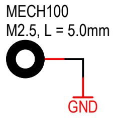
BOM:

M.2 connector (NGFF), B-Key:
H6.7 -> HOAUC, P/N = HYCW33B-05NGFF-670B
H6.7 -> Amphenol FCI, P/N = MDT670B02412

STAND-OFF HEIGHT TABLE

Connector Height Descriptor	L1
H2.3	0.35 ± 0.03
H2.5	0.55 ± 0.03
H2.8	0.80 ± 0.03
H3.2	1.45 ± 0.03
H4.2	2.45 ± 0.03

For 2230



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Mirkotronics
Aleksandry 3/89, 30-837 Kraków, POLAND

Size	A3
Title	PCB Marking & Testpoints
Version	VER2
Project:	UPiS Motherboard
Variant:	[No Variations]
Designer:	*
File Name:	[12] NGFF.SchDoc
RefDes:	
Sheet:	* / *
Printed:	3/9/2026
Revision	REV1